Payroll Settlement Module

1. Introduction

The Payroll Settlement module is one of the key components of the system, as it enables accurate calculation of employee salaries by applying deductions and adding benefits in accordance with Colombian regulations. This process ensures proper payroll payment and compliance with labor obligations.

2. Objectives

- Automate salary calculation by considering income and deductions.
- Facilitate the generation of pay stubs for each employee.
- Ensure accuracy in payroll settlement.
- Integrate results with other modules such as social benefits and reports.

3. Features

3.1 Net Salary Calculation

Determine salary based on:

- Days worked
- Bonuses and commissions

Apply deductions:

- Health (4%)
- Pension (4%)
- Withholding tax (if applicable)
- Other deductions (loans, garnishments, etc.)

3.2 Pay Stub Generation

Generate a document that includes:

- Breakdown of income and deductions
- Total amount payable
- Digital signature or verification code
- PDF format or direct print

4. Database Model

The following tables are required for this module:

Table: Payroll

Campo	Tipo de Dato	Descripción
id	INT (PK)	Identificador único
empleado_id	INT (FK)	Referencia al empleado
salario_base	DECIMAL(10,2)	Salario sin deducciones
deducciones	DECIMAL(10,2)	Total de deducciones
neto_pagar	DECIMAL(10,2)	Salario final tras descuentos
fecha_pago	DATE	Fecha de pago

Table: Deductions

Campo	Tipo de Dato	Descripción
id	INT (PK)	Identificador único
nomina_id	INT (FK)	Referencia a la nómina
tipo	VARCHAR(50)	Tipo de deducción
valor	DECIMAL(10,2)	Monto deducido

5. Processes and Algorithms

Net Salary Calculation

- Retrieve the employee's base salary
- Add bonuses
- Subtract mandatory deductions (health and pension)
- Apply additional deductions if applicable
- Store the result in the database

Pay Stub Generation

- Extract payroll information for the employee
- Format the data into a readable document

- Display it in the console

6. Conclusion

The Payroll Settlement module automates a key process in labor administration, ensuring that payments are accurate and comply with current regulations in Colombia. Its implementation reduces errors and optimizes management time.